Scope and Goals

- Deliver daily risk metrics
- Develop a backtest framework for strategies
- Provide execution analytics to reduce slippage and market impact

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Functional Requirements

1. **Daily Risk Analytics Job**
    - Compute Value at Risk (VaR) and Expected Shortfall (ES) at portfolio and strategy levels (95% and 99% confidence).
    - Provide factor exposure breakdowns and stress test results (rate shocks, volatility spikes, sector drawdowns).
    - Generate breach alerts with scenario evidence and impacted exposures.
2. **Backtesting Framework**
    - Support momentum and mean reversion strategies.
    - Include walk-forward validation, transaction cost modeling (commission, slippage, impact), position sizing, and risk constraints (max drawdown, volatility caps, sector/asset-class limits).
    - Ensure reproducibility by storing configs, seeds, versions, and outputs.
3. **Portfolio Optimization Routine**
    - Generate target weights under constraints (gross/net exposure, turnover, sector bounds, tracking error, risk budget).
    - Recommend trades to transition from current holdings.
    - Provide expected risk/return report using ex-ante factor and variance models.
4. **Transaction Cost Analysis (TCA) Module**
    - Measure realized slippage, market impact, and venue/algorithm performance.
    - Attribute P&L to alpha, beta, cost, and timing.
    - Deliver weekly TCA reports with trends and recommendations to reduce costs.
5. **Daily Risk & Performance Report**
    - Include: VaR/ES figures, factor contributions, drawdown, rolling Sharpe ratio, exposure summaries, TCA highlights, and exceptions.
    - Reports must be audit-ready, linking metrics to data snapshots, code versions, and configs.